

Fragmented but Not Flipped: A Controlled Re-Evaluation of Semantic-Communication Methods and a Minimal Viable Harness

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Abstract—Semantic-communication research evaluates methods with mutually incompatible harnesses: across a citation-ranked sample of 21 papers we find 31 distinct quality metrics (21 used by a single paper), an empty intersection of signal-to-noise-ratio ranges across image-domain papers, no preprocessing specified in 15 of 21, and the de-facto baseline DeepJSCC used explicitly by only 3 of 11 image papers. To test whether this fragmentation changes conclusions, we conduct a pre-registered controlled re-evaluation of two methods—a convolutional and a Swin-transformer joint source-channel coding system—under two harness configurations drawn verbatim from surveyed papers, one ranked by structural similarity and one by downstream classification accuracy. The ranking does not invert: the transformer dominates the convolutional method under both configurations, with margins well beyond the pre-registered noise floor. For this pair, the risk of fragmentation is therefore not silent rank inversion but unverifiable incomparability—the field cannot confirm whether such robustness holds generally. We propose and release a minimal viable evaluation harness that restores comparability and lets others test whether the result generalizes.

Index Terms—semantic communications, evaluation benchmark, reproducibility, joint source-channel coding, rank inversion, robustness

I. Introduction

Under a controlled common evaluation harness, the ranking of two semantic-communication (SemCom) methods is stable across two distinct harness configurations—one ranked by structural similarity (SSIM), the other by downstream classification accuracy: a Swin-transformer joint source-channel coding (JSCC) method dominates a convolutional one regardless of channel, SNR range, or metric. For this pair, the danger of evaluation fragmentation is therefore not silent rank inversion but unverifiable incomparability: with 31 distinct metrics across surveyed papers and no overlapping signal-to-noise-ratio (SNR) range, the field cannot confirm whether such robustness holds generally. We quantify this fragmentation across a citation-ranked sample of 21 papers: 31 distinct quality metrics (21 used by a single paper), an empty intersection of SNR ranges across image-domain papers, no preprocessing specified in 15 of 21, and DeepJSCC [1] used as an explicit baseline in only 3 of 11 image papers. We contribute: (i) a quantified fragmentation analysis;

(ii) a pre-registered controlled re-evaluation testing for rank inversion under harness variation; and (iii) a minimal viable evaluation harness.

II. Related Work

Semantic communication has converged on de-facto baselines—DeepSC [2] for text and DeepJSCC [1] for images—but not on a common evaluation harness around them. Existing benchmarks do not close this gap. 6G-Bench [3] evaluates foundation-model reasoning over network decision tasks rather than joint source-channel transmission, and is therefore orthogonal to image- and text-domain SemCom evaluation. Recent roadmaps list a unified evaluation methodology as a future requirement rather than delivering one. Our gap-killer search found no candidate satisfying all three criteria simultaneously: a fixed reusable transmission protocol, explicit adoption by five or more subsequent works, and publication after 2022 (see Section III for methodology). Survey papers assert this standardization gap as the field’s principal obstacle [4], [5] but propose no concrete protocol. This paper supplies the missing quantification of the gap, a demonstration of its consequences, and a minimal protocol to close it.

III. Fragmentation Analysis

We assembled a citation-ranked sample of SemCom papers from Semantic Scholar, stratified by year—the 15 most-cited for 2022–2023, the 10 most-cited for 2024, and the 10 most recent for 2025–2026—including only papers that report quantitative results with a specified evaluation setup, and excluding surveys and zero-result papers. Harness fields were extracted verbatim from each full text, yielding 21 papers with extractable harnesses from a 35-paper frame; the remaining 14 (10 inaccessible, 4 reporting no original evaluation harness) are preserved as labeled slots in the released table [6]. Table I shows six representative rows spanning the three strata. The same frame supplied the gap-killer search of Section II: no candidate satisfied a reusable protocol, five-or-more adopters, and post-2022 publication simultaneously.

The harness is fragmented on every axis. Metrics (F1): the 21 papers employ 31 distinct quality metrics; 21 appear in exactly one paper and only six in three or more. A single primary metric is explicitly declared in just 5 papers, while 13 report multiple metrics with no stated hierarchy and 3 optimize a system objective incommensurable with fidelity or task metrics. SNR (F2): reported SNR ranges across image-domain papers have an empty common intersection—their union spans $[-3, 21]$ dB, yet no single value is evaluated by all. Preprocessing (F3): 15 of 21 papers specify no preprocessing and only 2 specify it fully. Baselines (F4): the de-facto image baseline DeepJSCC is used as an explicit baseline in only 3 of 11 image-domain papers.

Harness variance alone is expected in a young field. Section IV tests whether it changes conclusions; for the pair examined, the ranking is stable, so the cost is incomparability rather than demonstrated inversion.

IV. Controlled Re-Evaluation Under Harness Variation

To test whether harness fragmentation changes conclusions, we re-evaluate two methods under two harness configurations drawn verbatim from surveyed papers. The experiment was pre-registered on 2026-06-07 (UTC), before any result was produced, fixing the hypotheses, both configurations, and the decision rule. We compare M1, the canonical CNN DeepJSCC [1], against M2, the Swin-transformer WITT [13], on a shared CIFAR-10 substrate.

Both methods transmit through a single shared channel module rather than their native implementations, which differ in noise convention and fading model. The shared channel applies identical per-symbol complex fading with perfect-CSI zero-forcing equalization and Rician $K = 1$, so any measured difference reflects the method or the harness, not an incidental channel implementation. Config A reproduces the harness of [14]: AWGN and Rayleigh channels, SNR 2–12 dB, ranked by SSIM. Config B reproduces [12]: AWGN, Rayleigh, and Rician channels, SNR 0–12 dB, ranked by top-1 classification accuracy on reconstructed images. Both configurations are taken cell-for-cell from rows of Table I and were registered before measurement; they therefore cannot have been chosen to manufacture the outcome, and differ only along axes that real papers vary.

A rank inversion is declared only if the winning method differs between configurations by more than the pre-registered noise floor—0.5 dB PSNR-equivalent or 2% absolute accuracy—over three seeds. The measured means (Table II) are: Config A SSIM 0.8321 ± 0.0013 (M1) vs. 0.8964 ± 0.0006 (M2); Config B accuracy 0.4359 ± 0.0140 (M1) vs. 0.6649 ± 0.0073 (M2). The ranking does not flip—M2 wins both—so the hypothesis is not confirmed. This is not an insensitive-test artifact: both between-method margins far exceed their floors (Config A: SSIM 0.0643 vs. 0.0020, PSNR 2.775 dB vs. 0.5; Config B: accuracy 0.229 vs. 0.0223). Scope is one method pair: the transformer’s

advantage over the CNN proved robust to the metric and channel axes varied here; whether it generalizes is what a standardized harness would let others test.

V. Proposed Harness Framework

The four fragmentation axes admit a direct remedy: a Minimal Viable Harness (MVH) that mandates exactly one specification per axis (Table III). MVH is deliberately minimal—five fields, not a heavyweight benchmark—so adoption costs little: a fixed dataset and channel set, a declared SNR grid, one declared primary metric with a mandatory secondary, a named baseline including DeepJSCC, and an explicit preprocessing specification. Each field is the cure for a quantified Section III finding rather than an arbitrary choice. Two modality profiles instantiate it: an image profile (CIFAR-10/Kodak; PSNR or SSIM primary with a perceptual secondary) and a text profile (BLEU primary with a semantic-similarity secondary). Reporting both a distortion and a task metric, as MVH requires, lets readers see at a glance whether a method’s advantage is metric-dependent—the question Section IV answers for one pair. We release a reference implementation—the shared channel and evaluation code—as a contribution, lowering adoption to a drop-in import.

VI. Conclusion

We quantified semantic-communication evaluation fragmentation across four axes—31 distinct metrics (21 used once), an empty SNR intersection across image-domain papers, preprocessing unspecified in 15 of 21 papers, and DeepJSCC an explicit baseline in only 3 of 11 (C1). A pre-registered controlled re-evaluation found no rank inversion for the tested pair: a transformer JSCC method outranked a CNN under both an SSIM- and an accuracy-based harness, with margins well beyond the noise floor (C2). We proposed and released a minimal viable harness that restores comparability (C3). This does not establish field-wide robustness—it tests one pair under one shared harness; the harness is what would let others test more pairs and verify whether this robustness generalizes. Limitations: one method pair, the image domain, and an analog-symbol channel approximation.

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TABLE I
Representative evaluation-harness fragmentation (6 of 21 papers; full table in supplement [6]).

Paper (year)	Domain	Channel	SNR (dB)	Primary metric	Baseline(s)	Preproc.
DeepJSCC-SemCom [7] ('22)	Image	AWGN	0–20	PSNR (+MS-SSIM, LPIPS, top-1)	BPG+polar, H.265	No
DeepSC-Speech [8] ('22)	Speech	AWGN+Rayleigh	5–10	WER, MCD	DeepSC-SR	Yes
Diffusion-JSCC [9] ('24)	Image	AWGN+Rayleigh+MIMO	3–12	PSNR/SSIM/LPIPS/FID/CLIP	CDDM	Partial
LLM-PHY [10] ('24)	PHY/CSI	MIMO-OFDM (AWGN)	5–20 / 0–25	NMSE	— (unnamed)	No
DSRDM [11] ('26)	Image	AWGN+Rician	0–20	BER	16/256-QAM	No
TONIC [12] ('26)	Image	AWGN+Rayleigh+Rician	0–12	Classification accuracy	DeepJSCC, JPEG	No
Aggregate ($n=21$)	—	—	$\cap = \emptyset$	31 distinct; 5 declared	DeepJSCC: 3/11	None: 15/21

TABLE II

Method ranking under two real-paper-derived harnesses (Config A: AWGN+Rayleigh, 2–12 dB; Config B: AWGN+Rayleigh+Rician, 0–12 dB). Mean over 3 seeds; bold marks the better method per column. Rank inversion NOT CONFIRMED: the winner (M2) does not flip, and both margins exceed the noise floor (0.5 dB PSNR-eq. / 2% acc.).

Method	Config A: SSIM $\uparrow$	Config B: Top-1 acc. $\uparrow$
M1 — DeepJSCC [1]	0.8321 $\pm$ 0.0013	0.4359 $\pm$ 0.0140
M2 — WITT [13]	0.8964 $\pm$ 0.0006	0.6649 $\pm$ 0.0073
Winner	M2	M2

TABLE III

Minimal Viable Harness: each mandated field cures a fragmentation finding.

Mandated field	Specification	Cures
Dataset & channel	Fixed public dataset; AWGN+Rayleigh (+Rician)	F2 + F4 (shared substrate)
SNR grid	Declared fixed dB test points	F2 (SNR $\cap = \emptyset$)
Primary + secondary metric	One declared primary + $\geq$ 1 mandatory secondary	F1 (31 metrics; 5 declared)
Baseline	DeepJSCC mandatory, config reported	F4 (3/11 image)
Preprocessing	Explicit normalization/resize/aug	F3 (15/21 none)

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